FREELANCE PORTFOLIO: AUTOMATED ETL PIPELINE

Specialization: Data Quality Engineering, Power Query (M Code), & Master Data Governance

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GitHub Repo: github.com/akbarsyafiq6-arch/Powerquery_Data_Cleaning_Portfolio

1. EXECUTIVE SUMMARY & DATA GOVERNANCE ARCHITECTURE

This portfolio report presents the automated end-to-end data sanitization and quality audit pipeline across **4,000 raw transactional records** spanning 4 core domains (Logistics, E-Commerce, HRIS Payroll, and B2B CRM Leads). The entire framework enforces a **3-Layer Data Governance Architecture** (*Raw Data → Defect Audit Log → Ready-to-Use Master Data*) built on **Power Query M Code**, guaranteeing zero premature data loss.

4,000	3,265	735	1-Click
TOTAL RAW ROWS	READY-USE (81.6%)	DEFECT LOG (18.4%)	REFRESH ETL PIPELINE

2. CONSOLIDATED DATA QUALITY AUDIT MATRIX (4 BUSINESS DOMAINS)

No	Case Study Domain	Raw Rows	Ready-Use (Valid)	Defect (Audit Log)	Defect Rate	Pipeline Status
1	**Logistics & Inventory**	1,000	736 (73.6%)	264 (26.4%)	26.4%	VERIFIED
2	**E-Commerce Sales**	1,000	867 (86.7%)	133 (13.3%)	13.3%	VERIFIED
3	**HRIS & Payroll Compliance**	1,000	821 (82.1%)	179 (17.9%)	17.9%	VERIFIED
4	**CRM B2B Lead Normalization**	1,000	841 (84.1%)	159 (15.9%)	15.9%	VERIFIED
TTL	**Compiled 4 ETL Pipelines**	**4,000**	**3,265 (81.6%)**	**735 (18.4%)**	**18.4%**	AUTOMATED

Access Pure M Code Scripts & Verified Workbooks:

All Excel working workbooks (`.xlsx`) and standalone modular M Code scripts (`.m`) are directly accessible and downloadable from the official public GitHub repository:
https://github.com/akbarsyafiq6-arch/Powerquery_Data_Cleaning_Portfolio.git

3. TECHNICAL CASE STUDY BREAKDOWN

Case Study 1: Logistics & Inventory Control Pipeline

DEFECT RATE: 26.4%

```
Dataset: /Datasets/01_Logistics_&_Inventory_Control_Pipeline.xlsx
M Code: /M_Scripts/01_Logistics_&_Inventory_Control_Pipeline.m
```

RAW DATA ISSUES

Negative stock quantities caused by entry typos, 'UNKNOWN' string unit prices, duplicate Txn_IDs, and non-standard warehouse zones.

M CODE TRANSFORMATIONS

Utilized `Number.Abs` for quantity recovery, isolated duplicate IDs to Defect log, parsed numeric unit price, and capitalized warehouse zones.

Business Impact: Prevents negative physical inventory records in Warehouse Management Systems (WMS) and eliminates double-counted stock valuation.

Case Study 2: End-to-End E-Commerce Sales Reconciliation

DEFECT RATE: 13.3%

```
Dataset: /Datasets/02_End-to-End_E-Commerce_Sales_Reconciliation_Pipeline_in_Power_Query_M_Code.xlsx
M Code: /M_Scripts/02_End-to-End_E-Commerce_Sales_Reconciliation_Pipeline_in_Power_Query_M_Code.m
```

RAW DATA ISSUES

Net Amount polluted with 'Rp' currency symbols, mixed string/negative values, duplicate Order_IDs, and unformatted order dates.

M CODE TRANSFORMATIONS

Extracted non-numeric characters, standardized payment channels (CAPITAL), and routed valid transactions to ReadyUse and anomalies to Defect log.

Business Impact: Delivers accurate net revenue reporting for monthly financial reconciliation without distortion from bad records.

Case Study 3: HRIS Master Data & Payroll Compliance Cleaner

DEFECT RATE: 17.9%

```
Dataset: /Datasets/03_Automated_HRIS_Master_Data_Audit_&_Payroll_Compliance_Cleaner.xlsx
M Code: /M_Scripts/03_Automated_HRIS_Master_Data_Audit_&_Payroll_Compliance_Cleaner.m
```

RAW DATA ISSUES

Invalid national ID numbers (NIK < 16 digits or 'INVALID_NIK'), trailing spaces in employee names, and string overtime hours ('30 hours').

M CODE TRANSFORMATIONS

Strict 16-digit character length testing (`Text.Length = 16`), routing 13-digit NIKs to Defect log, numeric extraction, and space trimming.

Business Impact: Guarantees tax (PPh 21) and social security compliance by ensuring 100% of processed payroll records contain verified 16-digit NIKs.

Case Study 4: CRM B2B Lead Database Normalization

DEFECT RATE: 15.9%

```
Dataset: /Datasets/04_CRM_B2B_Lead_Database_Normalization_&_Anomaly_Detection_System.xlsx
M Code: /M_Scripts/04_CRM_B2B_Lead_Database_Normalization_&_Anomaly_Detection_System.m
```

RAW DATA ISSUES

Corrupted email addresses ('AT' replacing '@', domain suffix '.co.iddd'), negative Lead Scores, and risk of truncated leading '0's in phone numbers.

M CODE TRANSFORMATIONS

Reconstructed email strings, enforced Text data type on Phone Numbers to preserve '08...' prefix, and filtered out invalid Lead Scores.

Business Impact: Boosts B2B email deliverability and ensures contact phone numbers are ready for automated WhatsApp Gateway / Telemarketing outreach.

4. GITHUB REPOSITORY ARCHITECTURE & KEY M CODE SNIPPET

The GitHub repository directory is organized following modern software engineering conventions for seamless technical review:

```
Powerquery_Data_Cleaning_Portfolio/ ├── README.md # Main documentation & M Code summary ├── /Docs/ # Print-ready PDF Portfolio Reports │ ├── portofolio_freelance_data_cleaning.pdf │ └── freelance_data_cleaning_portfolio_en.pdf ├── /Datasets/ # Processed Excel Workbooks (.xlsx) │ ├── 01_Logistics_&_Inventory_Control_Pipeline.xlsx │ ├── 02_End-to-End_E-Commerce_Sales_Reconciliation.xlsx │ ├── 03_Automated_HRIS_Master_Data_Audit_&_Payroll.xlsx │ └── 04_CRM_B2B_Lead_Database_Normalization.xlsx └── /M_Scripts/ # Modular Pure M Code Files (.m) ├── 01_Logistics_&_Inventory_Control_Pipeline.m ├── 02_End-to-End_E-Commerce_Sales_Reconciliation.m ├── 03_Automated_HRIS_Master_Data_Audit.m └── 04_CRM_B2B_Lead_Database_Normalization.m
```

M Code Highlight: Audit Information Details (HRIS)

```
let ID_Defect = if [Employee_ID] = null or Text.From(Text.Trim([Employee_ID])) = "" or [Employee_ID] <> null and [Count_ID] > 1 then "Defect/Dupicates ID;" else "", NIK_Defect = if [NIK_KTP_Clean] = null or Text.From(Text.Trim([NIK_KTP_Clean])) = "" or [NIK_KTP_Clean] <> null and [Count_NIK] > 1 or Text.Length (Text.From([NIK_KTP_Clean])) <> 16 then "Defect/Dupicates NIK;" else "", Base_Salary_Defect = if [Base_Salary_IDR] = null then "Defect Base Salery;" else "", Overtime_Defect = if [Overtime_Hours_Clean] = null or [Overtime_Hours] < 0 then "Defect Overtime" else "", Join_Date_Defect = if [Join_Date] = null then "Defect Join Date" else "" in ID_Defect & NIK_Defect & Base_Salary_Defect & Overtime_Defect & Join_Date_Defect
```

5. FREELANCE SERVICES & CONSULTING OFFERINGS

Are Dirty Data & Manual Spreadsheet Tasks Slowing Down Your Business?

I provide professional data engineering services to design 1-Click Refresh ETL cleaning pipelines, isolate duplicates, and audit database quality across Power Query Excel & Power BI environments.

Core Freelance Services:

- Automated Data Cleaning & ETL Pipelines
- 3-Layer Data Quality Audit Log System
- Master Data Normalization (HRIS/CRM/Sales)
- Power BI & Excel Executive Dashboarding

Contact & Portfolio:

Email: akbarsyafiq6@gmail.com
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PDF Portfolio: Available in `/Docs/` Folder